



IDENTIFYING OUTLIERS IN RETAIL SALES DATA

FOR BETTER BUSINESS DECISIONS

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1 INTRODUCTION

Retail datasets may contain unusually high or low sales values. These observations can influence averages, trends and forecasts. Outlier analysis helps separate unusual events.

2 OBJECTIVES

Detect unusual sales transactions
Measure their influence on analysis.
Identify possible causes
Improve forecasting , inventory planning

3 DATASET OVERVIEW

Dataset: Retail Sales Data
Features: Date, Product, Category,
Quantity, Unit Price, Total Sales
Main variable: Sales Amount
Possible outliers: bulk orders

4 WHY OUTLIERS MATTER

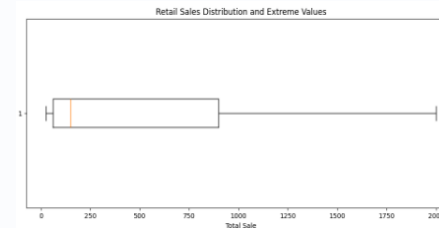
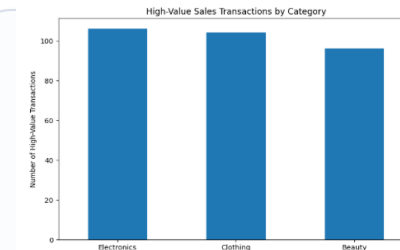
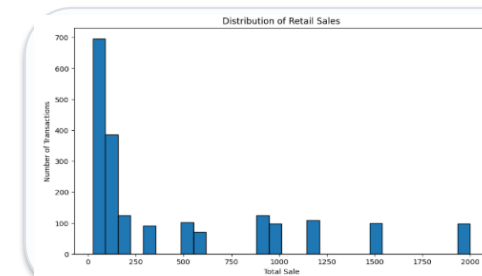
Extreme values may:

- Distort the mean and standard deviation
- Affect demand forecasts
- Mislead performance comparisons

5 METHODOLOGY



6 EXPLORATORY DATA ANALYSIS & VISUALIZATION



7 KEY FINDINGS

- Outliers can strongly change summary statistics.
- A high value may be genuine business activity.

8 BUSINESS IMPACT

- Better demand forecasting
- Smarter inventory planning
- More reliable KPIs
- Improved anomaly monitoring
- Reduced risk of misleading

9 TOOLS & TECHNOLOGIES

Python
Pandas • NumPy
Matplotlib • Seaborn
Jupyter Notebook

10 CONCLUSION

Outlier detection is an important step in retail analytics. The best approach is to detect unusual values, investigate their cause, validate them and then decide whether to retain, correct or exclude them.